

IPC144 – Introduction to C

Programming

Practice Exercises

Week 2

Q1 - (Arithmetic) Write a program that asks the user to enter two numbers, obtains the two numbers from the user and prints the sum, product, difference, and quotient of the two numbers.

Q2 - (Final Velocity) Write a program than asks the user to enter the initial velocity and acceleration of an object, and the time that has elapsed, places them in the variables u, a, and t, and prints the final velocity, v, and distance traversed, s, using the following equations.

$$a) v = u + at$$

$$b) s = ut + \frac{1}{2} at^2$$

Q3 - (Comparing Integers) Write a program that asks the user to enter the highest rainfall ever in one season for a country, and the rainfall in the current year for that country, obtains the values from the user, checks if the current rainfall exceed the highest rainfall and prints an appropriate message on the screen. If the current rainfall is higher, it assigns that value as the highest rainfall ever. Use only the single-selection form of the if statement you learned in this chapter.

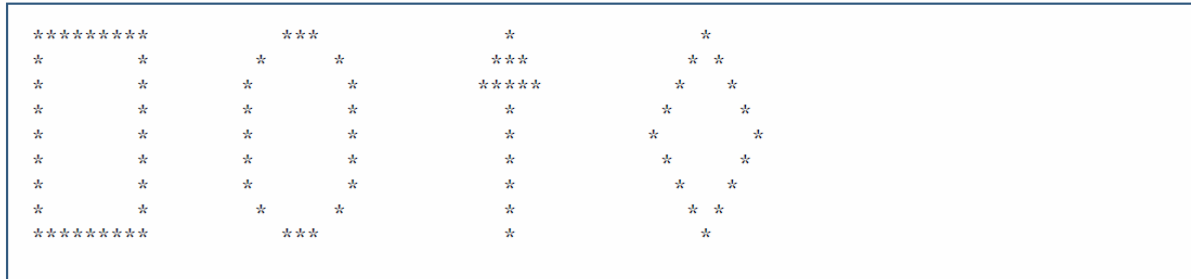
Q4 - (Arithmetic, Smallest and Largest) Write a program that inputs three integers from the keyboard and prints the sum, average, product, smallest and largest of these numbers. The screen dialog should appear as follows:

```
Enter three different integers: 13 27 14
Sum is 54
Average is 18
Product is 4914
Smallest is 13
Largest is 27
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Q5 - (Diameter, Circumference and Area of a Circle) Write a program that reads in the radius of a circle as an integer and prints the circle's diameter, circumference and area. Use the constant value 3.14159 for π . Do all calculations in output statements

Q6 – (Converting from seconds to hours, minutes and seconds) Write a program that asks the user to enter the total time elapsed, in seconds, since an event and converts the time to hours, minutes and seconds. The time should be displayed as hours:minutes:seconds. [Hint: Use the remainder operator]

Q7 - (Displaying Shapes with Asterisks) Write a program that prints the following shapes with asterisks.



Q8 - (Largest and Smallest Integers) Write a program that reads in 3 integers and determines and prints the largest and the smallest integers in the group.

Q9 - (Odd or Even) Write a program that reads an integer and determines and prints whether it's odd or even. [Hint: Use the modulus operator. An even number is a multiple of two. Any multiple of two leaves a remainder of zero when divided by 2.]

Q10 - (Multiples) Write a program that reads in two integers and determines and prints if the first is a multiple of the second. [Hint: Use the modulus operator.]

Q11 - (Integer Value of a Character) Here's a peek ahead. In this chapter you learned about integers and the type int. C can also represent uppercase letters, lowercase letters and a considerable variety of special symbols. C uses small integers internally to represent each different character. The set of characters a computer uses together with the corresponding integer representations for those characters is called that computer's character set. You can print the integer equivalent of uppercase A, for example, by executing the statement

```
printf( "%d", 'A' );
```

Write a C program that prints the integer equivalents of some uppercase letters, lowercase letters, digits and special symbols. As a minimum, determine the integer equivalents of the following: A B C a b c 0 1 2 \$ * + / and the blank character.

Q12 - (Separating Digits in an Integer) Write a program that inputs one five-digit number, separates the number into its individual digits and prints the digits separated from one

another by three spaces each. [Hint: Use combinations of integer division and the remainder operation.] For example, if the user types in 42139, the program should print

4 2 1 3 9

Q13 - (Table of Squares and Cubes) Using only the techniques you learned in this chapter, write a program that calculates the squares and cubes of the numbers from 0 to 10 and uses tabs to print the following table of values:

number	square	cube
0	0	0
1	1	1
2	4	8
3	9	27
4	16	64
5	25	125
6	36	216
7	49	343
8	64	512
9	81	729
10	100	1000

Q14 - (Q15 - (Body Mass Index Calculator))

The formulas for calculating BMI are

$$\text{BMI} = \frac{\text{weightInPounds} \times 703}{\text{heightInInches} \times \text{heightInInches}}$$

$$\text{BMI} = \frac{\text{weightInKilograms}}{\text{heightInMeters} \times \text{heightInMeters}}$$

Create a BMI calculator application that reads the user's weight in pounds and height in inches (or, if you prefer, the user's weight in kilograms and height in meters), then calculates and displays the user's body mass index. Also, the application should display the following information from the Department of Health and Human Services/National Institutes of Health so the user can evaluate his/her BMI:

BMI VALUES

Underweight: less than 18.5

Normal: between 18.5 and 24.9

Overweight: between 25 and 29.9

Obese: 30 or greater

Q15 - (*Car-Pool Savings Calculator*) Create an application that calculates your daily driving cost, so that you can estimate how much money could be saved by carpooling, which also has other advantages such as reducing carbon emissions and reducing traffic congestion. The application should input the following information and display the user's cost per day of driving to work:

- a) Total miles driven per day.
- b) Cost per gallon of gasoline.
- c) Average miles per gallon.
- d) Parking fees per day.
- e) Tolls per day.